Machine Learning for Customer Churn Prediction

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1 Executive Summary

Customer Churn is a problem about customer discontinue a company's services or products. Machine learning techniques can be applied with purpose of predict if a customer has probability of be churn, this information can help the company to take decisions for not lost clients. This work search explore different machine learning techniques to compare and get the best approach to solve this problem.

2 Dataset Download and Integrity Verification

2.1 Dataset Source and Acquisition

The dataset is the **WSDM – KKBox Music Service Churn Prediction Challenge**, publicly available through the Kaggle competition platform at
`https://www.kaggle.com/competitions/kkbox-churn-prediction-challenge`.
The dataset was released by KKBox Co., Ltd. under the competition's terms of use, which permit academic and non-commercial research purposes. No paid subscription is required; a free Kaggle account is sufficient to access all files. To reproduce this experiment, the following two steps are required:

1. **Accept the competition terms:** Create a free Kaggle account, navigate to the competition page, and accept the data usage rules. This step is mandatory before any file can be downloaded.
2. **Download and place the raw files:** Download all nine `.csv.7z` archives provided by the competition and place them, **without extracting**, in the directory `workshops/data/`. The preprocessing pipeline extracts and converts them automatically during execution.

No further manual setup is required. The data pipeline reads the compressed archives directly from `workshops/data/` and writes the intermediate `.parquet` files to `workshops/data/parquet/`. The final processed cohorts are produced in `workshops/w1/export-tabular-v5/`.

2.2 Dataset Characteristics

The competition provides nine compressed files (`.csv.7z`). Table 1 summarises each file, its role in the pipeline, and its approximate uncompressed row count.

Table 1: KKBox dataset files and their characteristics.

File	Description	Rows (approx.)	Columns
`train.csv`	February cohort labels: user ID and churn target	992,931	2
`train_v2.csv`	March cohort labels: user ID and churn target	970,960	2
`members_v3.csv`	User demographic profiles (age, gender, city, etc.)	6,769,473	7
`transactions.csv`	Earlier subscription transaction records	19,435,022	9
`transactions_v2.csv`	Recent subscription transaction records	21,591,521	9
`user_logs.csv`	Earlier daily listening behaviour logs	18,396,362	9
`user_logs_v2.csv`	Recent daily listening behaviour logs	18,396,362	9
`sample_submission_v2.csv`	Test set user IDs for competition submission	224,000	2
`sample_submission_zero.csv`	Baseline submission with all predictions set to zero	224,000	2

2.3 Integrity Verification

The Kaggle platform does not publish official checksums for competition datasets. To guarantee that collaborators download the correct files, SHA-256 hashes were computed directly on the nine `.csv.7z` archives using the project script `workshops/w1/src/data_info/dataset_info.py`. The full hashes are stored in `workshops/data/checksums.json`; Table 2 shows the first 16 characters of each digest together with the compressed file size on disk.

Table 2: SHA-256 checksums of the raw `.csv.7z` archives (first 16 hex characters shown).

File	SHA-256 (prefix)	Size (MB)
train.csv.7z	e2e7cec3392158ad	33.6
train_v2.csv.7z	683158156862e226	32.8
members_v3.csv.7z	a0e0b92b2ef9cc11	242.3
transactions.csv.7z	ff273f50f8e690c7	707.5
transactions_v2.csv.7z	78d01c8e90b86479	48.9
user_logs.csv.7z	2d21a83ddeeaf508	7,136.1
user_logs_v2.csv.7z	5e7b1d2b6fce3ff0	686.0
sample_submission_v2.csv.7z	329dfe694ccc76c7	30.7
sample_submission_zero.csv.7z	8d709c30a1e7dc52	32.8

To verify the downloaded files, run the following command from the project root:

```
python workshops/w1/src/data_info/dataset_info.py
```

The script computes the SHA-256 of each archive in `workshops/data/`, compares the result against `checksums.json`, and prints `OK` or `FAIL` for each file. It exits with code 1 if any file is missing or does not match.